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Pose Detection Accuracy Analysis Using Media Pipe(Member-2)

Project Overview

Human pose detection is a computer vision technique used to identify and track key body landmarks from images or videos. In this project, Media Pipe Pose was used to detect human body landmarks and analyse the accuracy of detection under different conditions such as front view, side view, low lighting, and different body poses. The study helps evaluate the effectiveness of pose estimation models for applications such as virtual try-on systems, fitness tracking, and human activity recognition.

Tools Used

- Python
- MediaPipe Pose
- OpenCV
- Matplotlib

Methodology

Step 1: Image Selection

Different images were selected representing various test conditions:

- Front View
- Side View
- Low Lighting
- Different Body Pose

Step 2: Pose Detection

MediaPipe Pose was applied to each image to detect human body landmarks.

Step 3: Landmark Visualization

The detected landmarks and skeletal connections were drawn on the image for visualization.

Step 4: Confidence Score Calculation

The average confidence score of all detected landmarks was calculated to evaluate detection quality.

Step 5: Performance Analysis

The confidence scores and output images were compared to analyse model performance under different conditions.

Experimental Results

Front View Image

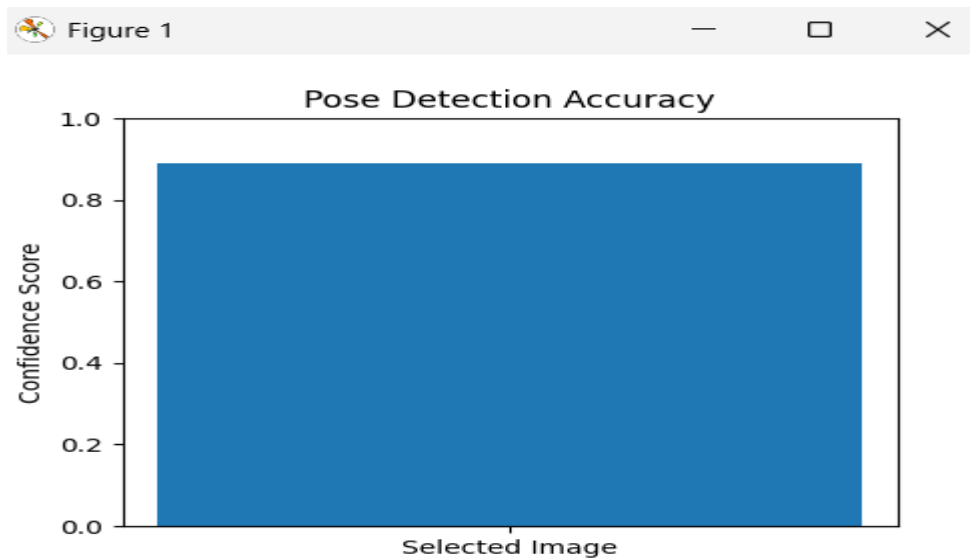
The model successfully detected most body landmarks with high accuracy. The complete skeletal structure was generated correctly.

Confidence Score: 0.89

Observation:

- Accurate landmark detection.
- Clear body structure identification.
- High overall performance.





Side View Image

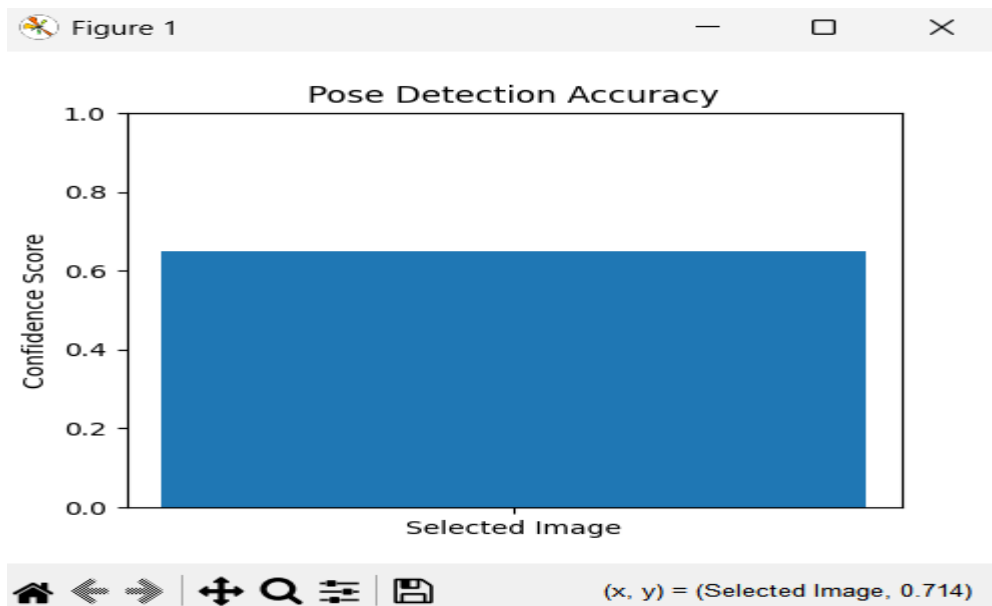
The side-view image reduced visibility of certain landmarks because some body parts were partially hidden.

Confidence Score: 0.71

Observation:

- Some landmarks became less visible.
- Detection performance slightly decreased.
- Skeleton generation remained stable.





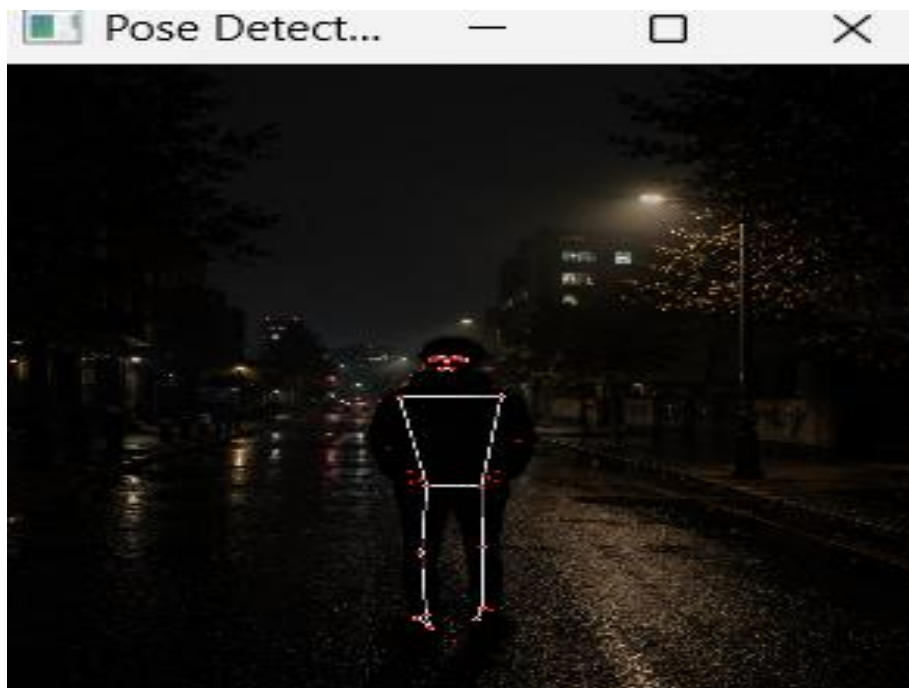
Low Lighting Image

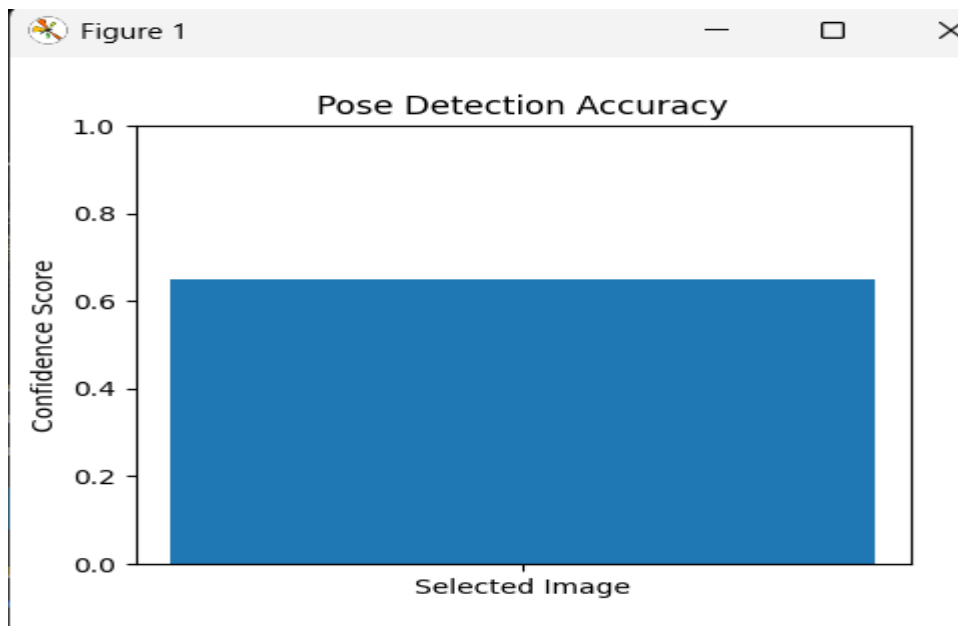
The model was tested using an image captured in poor lighting conditions.

Confidence Score: 0.65

Observation:

- Reduced visibility affected detection quality.
- Landmark confidence decreased.
- Human body was still detected successfully





Different Body Pose

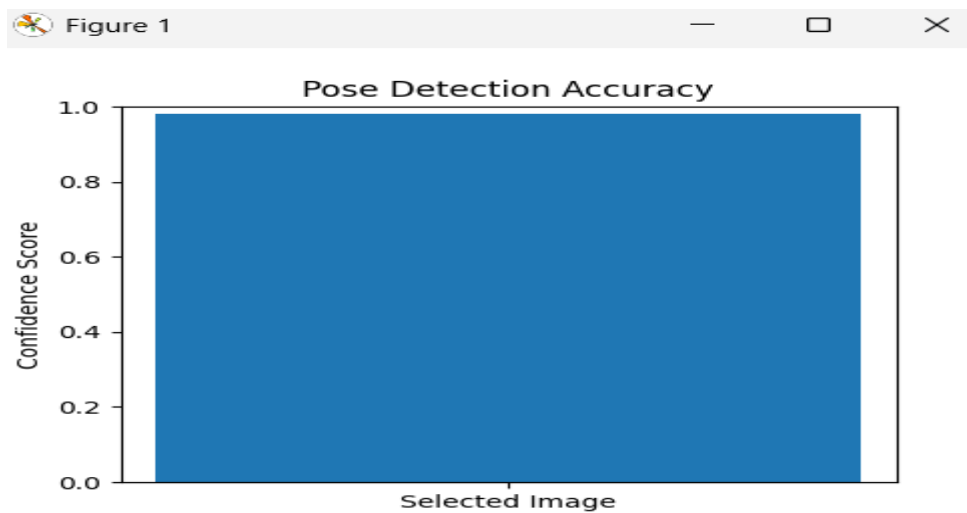
The model was tested using a different body posture.

Confidence Score: 0.98

Observation:

- Most landmarks were detected correctly.
- Excellent pose estimation performance.
- High confidence score achieved.





Accuracy Comparison Table

Condition	Confidence Score	Performance
Front View	0.89	High
Side View	0.71	Moderate
Low Lighting	0.65	Moderate
Different Pose	0.98	Very High

Accuracy Observations

- Front-view images achieved high detection accuracy.
- Side-view images slightly reduced landmark visibility.
- Low-light conditions affected overall confidence scores.
- Different poses were handled effectively by the model.
- Media Pipe Pose showed strong performance across multiple scenarios

Challenges Faced

- Landmark visibility decreased in side-view images.
- Poor lighting reduced confidence scores.
- Occluded body parts affected detection quality.
- Image quality influenced overall performance.

My Observation

The experiments demonstrated that Media Pipe Pose can accurately detect human body landmarks under various conditions. The highest performance was observed for images with clear visibility and favourable lighting. Although side views and low-light environments reduced detection confidence, the model consistently generated reliable pose estimations. Overall, Media Pipe Pose is an effective and efficient solution for pose detection tasks and can be successfully applied in virtual try-on systems and other computer vision applications.